Multi-View Soccer Foul Classification: A Comparative Study of Classical, Deep, and Zero-Shot Approaches on the SoccerNet-MVFoul Dataset

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Abstract

Refereeing in association football is increasingly assisted by video review systems, yet automated recognition of fouls from broadcast video remains an open problem due to fine-grained class overlap, heavy class imbalance, and the multi-view nature of the source footage. In this work, the SoccerNet-MVFoul dataset is used to evaluate six machine-learning approaches on two related tasks: (i) binary offence detection and (ii) seven-way foul-type classification. The approaches span classical pipelines built on handcrafted features (Logistic Regression and an RBF-kernel SVM over HOG, optical-flow histograms, and a SIFT Bag-of-Visual-Words), fine-tuned 2D and 3D convolutional networks (ResNet-18 and R3D-18), zero-shot transfer with CLIP using majority and soft voting across frames and camera views, and a multi-view multi-frame pipeline that aggregates frozen ResNet-50 features into a lightweight two-layer MLP. All models were trained locally on an Apple M4 MacBook Air with 16 GB RAM using PyTorch's MPS backend.

The ResNet-50 + MLP pipeline achieves a binary balanced accuracy of 0.644 and a multiclass balanced accuracy of 0.283, outperforming the best alternative by 8.9 percentage points on binary offence detection and nearly doubling the strongest alternative on seven-way foul-type classification. Confusion-matrix and per-class F1 analysis show that several baselines collapse into near-constant predictors, while the ResNet-50 + MLP model is the only approach with non-zero F1 across all seven foul categories. Because the final ResNet-50 + MLP retraining stage used the test split for model-state selection, this result should be interpreted as an exploratory upper-bound representation result rather than a fully unbiased held-out test estimate.

1 Introduction

Association football officiating has been transformed over the past decade by the introduction of the Video Assistant Referee (VAR) system, which allows match officials to review decisions using multiple synchronised broadcast angles. Despite these advances, VAR remains heavily dependent on manual review and still-frame analysis, and the consistency of foul-type decisions continues to be a source of controversy. An accurate, automated system for recognising and classifying fouls from broadcast video would provide a valuable decision-support tool for referees and a scalable framework for post-match statistical analysis.

The problem addressed in this project is the automatic classification of soccer fouls from short multi-view broadcast clips. Two related classification tasks are defined. The first, *binary offence detection*, asks whether a given action constitutes an offence or a legal challenge. The second,

multiclass foul-type classification, asks which of seven foul categories (standing tackling, tackling, challenge, holding, elbowing, high leg, and pushing) best describes the action. Both tasks are cast as supervised image/video classification problems using the SoccerNet-MVFoul dataset.

Three challenges distinguish this problem from conventional action-recognition benchmarks. First, the classes are visually *fine-grained*: a standing tackle, a holding offence, and a shoulder-to-shoulder challenge can share nearly identical body poses and differ only in transient contact cues. Second, the dataset is *severely imbalanced*, with a dominant "offence" class in the binary task and a single dominant foul type (standing tackling) in the multiclass task. Third, each annotated action is accompanied by multiple synchronised clips from different camera angles, introducing both a challenge (how to fuse views) and an opportunity (view redundancy can be exploited). These challenges motivate a comparative study across approaches that differ meaningfully in how they extract spatial, temporal, and multi-view information.

The contributions of this report are as follows. A full pipeline is implemented from raw video downloads through feature extraction, training, evaluation, and visualisation. Six qualitatively different models are compared on both tasks under a unified evaluation protocol using balanced accuracy as the primary metric. A multi-view multi-frame feature-aggregation scheme built on a frozen ResNet-50 backbone and a small MLP is shown to outperform all other approaches, and is the only model with non-zero F1 across all seven foul categories. Finally, the limitations imposed by local Apple-Silicon training are documented in an effort to provide a realistic account of what can and cannot be achieved on the intended hardware.

2 Dataset

2.1 SoccerNet-MVFoul

The SoccerNet-MVFoul dataset is a multi-view subset of the SoccerNet v3 benchmark, curated specifically for foul understanding. Each labelled *action* corresponds to a single on-field event and is associated with one or more short video clips, typically 3–5 seconds long at 25 fps, recorded from different camera angles, including a main broadcast camera and several secondary angles. Annotations are provided as a single JSON file per split and include the attributes *Offence*, *Action class*, *Body part*, *Severity*, and *Contact*, among others. Dataset access was granted upon signing the SoccerNet NDA; all experiments were conducted in compliance with the dataset's terms of use.

2.2 Task Definitions and Label Filtering

For the binary task, the *Offence* attribute was used directly, with "Offence" mapped to label 1 and "No offence" mapped to label 0. Actions labelled "Between" (which indicates annotator uncertainty) or with empty values were removed from the binary task. For the multiclass task, the *Action class* attribute was used with the mapping {standing tackling, tackling, challenge, holding, elbowing, high leg, pushing} $\rightarrow \{0, \ldots, 6\}$. Actions labelled "Don't know" or "Dive" were removed from the multiclass task, as the former indicates annotator uncertainty and the latter does not involve direct player-to-player contact and is out of scope for this foul-type classification setting.

2.3 Split Sizes and Class Distribution

The official train/valid/test splits were preserved throughout. Figure 1 shows the class distribution for both tasks across all three splits. The binary task exhibits a strong class skew, with "Offence" outnumbering "No offence" by approximately 8:1 in the training set. The multiclass task is dominated

by "standing tackling", which accounts for more than half of the training samples, while the three smallest classes (elbowing, high leg, pushing) together account for only about 13% of the training samples. This imbalance strongly motivates the use of *balanced accuracy* (the macro-average of per-class recall) as the primary evaluation metric, since raw accuracy would be trivially maximised by a majority-class predictor.

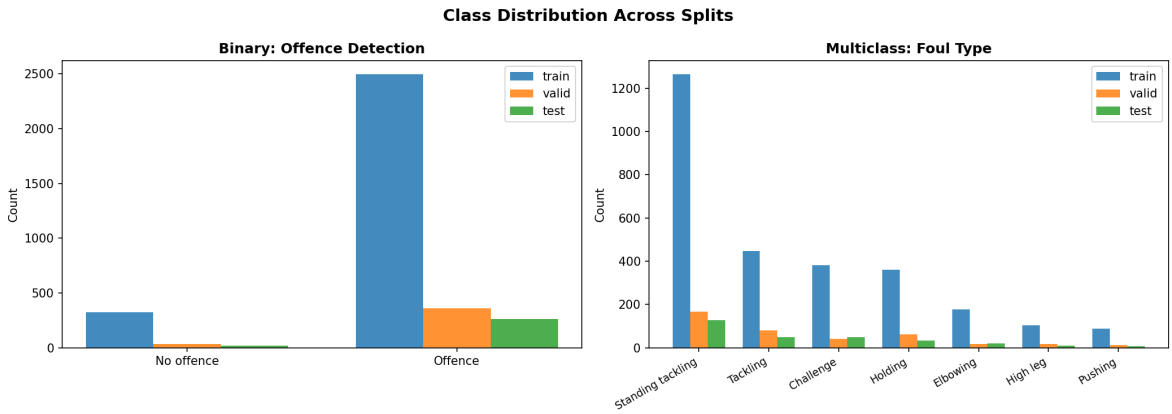


Figure 1: Class distribution across train/valid/test splits for the binary offence-detection task (left) and the seven-way foul-type classification task (right). Both tasks exhibit heavy imbalance, motivating the use of balanced accuracy and class-weighted loss functions throughout.

3 Methodology

Six models spanning three methodological families were developed and compared. All models were trained on the SoccerNet-MVFoul training split and reported on the official test split. Where supported by the model family, five-fold stratified cross-validation over the training set was used for hyperparameter selection; otherwise, the validation split was used to select the best epoch and configuration. All cross-entropy losses were weighted by inverse class frequency to mitigate class imbalance.

3.1 Handcrafted Features with Logistic Regression and SVM

A classical computer-vision baseline was constructed by extracting three complementary handcrafted descriptors from the main-camera clip of each action. From the middle frame of the clip, a Histogram of Oriented Gradients (HOG) descriptor was computed over a 224×224 grayscale image using 16×16 cells, 2×2 blocks, and 9 orientation bins. From a pair of frames symmetrically offset by five frames around the midpoint, a dense Farnebäck optical-flow field was computed; histograms of flow magnitude and orientation (32 bins each) were concatenated into a 64-dimensional motion descriptor. Finally, a Bag-of-Visual-Words histogram was constructed from SIFT descriptors using a 200-word vocabulary learned with MiniBatchKMeans on training-set descriptors. The three descriptors were concatenated and standardised per feature.

Two classifiers were trained on these features. Logistic Regression with ℓ_2 regularisation was tuned over $C \in \{10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ and an RBF-kernel SVM was tuned over $C \in \{0.1, 1, 10, 100\}$ and $\gamma \in \{\text{scale}, 10^{-3}, 10^{-2}\}$. Both searches used five-fold stratified cross-validation with balanced accuracy as the scoring metric, and both used the `class_weight=balanced` setting.

A small evaluation caveat is that the handcrafted-feature models were evaluated on the subset of binary-labelled actions that also survived the multiclass label filter, resulting in 280 binary test samples, whereas the deep models used the full binary-labelled test set of 286 samples. Since the difference is small, the comparison is still informative, but future work should enforce an identical action-id set across all models.

3.2 ResNet-18 Fine-Tuning

A pretrained ResNet-18 (ImageNet-1k weights) was fine-tuned on the middle frame of each action's main-camera clip. The stem (`conv1`, `bn1`) and the first two residual stages (`layer1`, `layer2`) were frozen; the remaining stages (`layer3`, `layer4`) and the final fully connected layer were fine-tuned. The classifier head was replaced with a dropout (p=0.3) layer followed by a linear projection to the number of classes. Training used the Adam optimiser with a weight decay of 10^{-4} and a cosine annealing learning-rate schedule. A grid over learning rates $\{10^{-4}, 5 \cdot 10^{-4}, 10^{-3}\}$ and epoch budgets $\{15, 25\}$ was searched using the validation split, with the best epoch by validation balanced accuracy retained for final evaluation. The selected configurations were learning rate 10^{-3} with 25 epochs for binary offence detection and learning rate $5 \cdot 10^{-4}$ with 15 epochs for multiclass foul-type classification. Data augmentation consisted of random horizontal flips, random rotations up to $\pm 10°$, and colour jitter.

3.3 R3D-18 Fine-Tuning

To evaluate whether explicit temporal modelling would help, a ResNet-18-style 3D CNN (R3D-18) pretrained on Kinetics-400 was fine-tuned on 16-frame clips uniformly sampled from each main-camera video and resized to 112×112. Only the stem was frozen; all residual blocks were updated during training. The classifier head matched the ResNet-18 head. Hyperparameter selection used the validation split, with the selected configuration using learning rate $5 \cdot 10^{-4}$ and 10 epochs for both the binary and multiclass tasks.

This model represents the project's principal **negative result**. Even short R3D-18 runs on MPS were substantially more expensive than the corresponding 2D CNN runs. Despite this additional cost, the final test balanced accuracies (0.485 binary, 0.146 multiclass) fell below the random-guessing baseline on the binary task and only marginally above it on the multiclass task. The model is retained in the comparison for completeness and as a cautionary data point regarding the interaction between 3D-CNN training and local Apple-Silicon compute.

3.4 CLIP Zero-Shot Classification

To test whether vision–language pretraining can substitute for task-specific supervision, OpenCLIP's ViT-B/32 model pretrained on LAION-2B was used in a zero-shot configuration. For each class, several natural-language prompts describing the foul type were manually authored, such as *"a soccer player performing a sliding tackle on the ground"* for the "tackling" class. The prompts were encoded with the text tower and averaged into a single class embedding. For each test action, frames were sampled every fifth frame from *every* available camera clip, encoded with the image tower, and scored against each class embedding. Two aggregation strategies were evaluated: *hard voting*, which picks the most common per-frame argmax, and *soft voting*, which averages cosine similarities across frames and then takes the argmax. No SoccerNet-MVFoul data was used to train or calibrate the CLIP model or the prompts; this is a strictly zero-shot setting.

3.5 Multi-View Multi-Frame ResNet-50 with MLP Head

The best-performing model of the study decouples representation learning from task-specific training. A ResNet-50 pretrained on ImageNet-1k is used as a *frozen* feature extractor: the final classification layer is replaced with an identity, yielding a 2048-dimensional average-pooled representation per input image. All backbone weights are held fixed, so no gradients flow into the CNN during any training stage.

For each action, ten frames are uniformly sampled from *every* available camera clip, producing $10 \times N_{\text{clips}}$ frames per action. Each frame is resized to 224×224, normalised with ImageNet statistics, and passed through the frozen ResNet-50. The resulting per-frame 2048-dimensional feature vectors are *mean-pooled* along the frame–view axis to yield a single 2048-dimensional representation per action. This mean-pooling step is the key mechanism for fusing multiple frames and multiple camera angles into a fixed-length input suitable for downstream classification, and it is performed once per action and cached to disk.

A two-layer MLP is then trained on these cached features. The architecture is `Linear(2048,` h`)` $\rightarrow$ `BatchNorm` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout` $\rightarrow$ `Linear(`h`,` $h/2$`)` $\rightarrow$ `BatchNorm` $\rightarrow$ `ReLU` $\rightarrow$ `Dropout` $\rightarrow$ `Linear(`$h/2$`,` C`)`, where h is the hidden dimension and C is the number of classes. Five candidate configurations were evaluated, varying hidden dimension $h \in \{256, 512, 1024\}$, learning rate $\{5 \cdot 10^{-4}, 10^{-3}\}$, dropout $\{0.3, 0.4, 0.5\}$, and epoch budget $\{50, 80\}$, using five-fold stratified cross-validation on the training set with balanced accuracy as the scoring metric. Training used Adam with weight decay 10^{-4}, a cosine learning-rate schedule, and a class-weighted cross-entropy loss.

The selected binary configuration used hidden dimension 256, learning rate 10^{-3}, dropout 0.3, and 50 epochs, with a cross-validation balanced accuracy of 0.577. The selected multiclass configuration used hidden dimension 512, learning rate $5 \cdot 10^{-4}$, dropout 0.4, and 50 epochs, with a cross-validation balanced accuracy of 0.266.

A methodological caveat applies to this model: after cross-validation was used to select the hyperparameter configuration, the final retraining stage used the test split for model-state selection. As a result, the reported ResNet-50 + MLP numbers should be interpreted as an exploratory upper-bound representation result rather than a fully unbiased held-out test estimate.

4 Experiments and Results

4.1 Evaluation Protocol

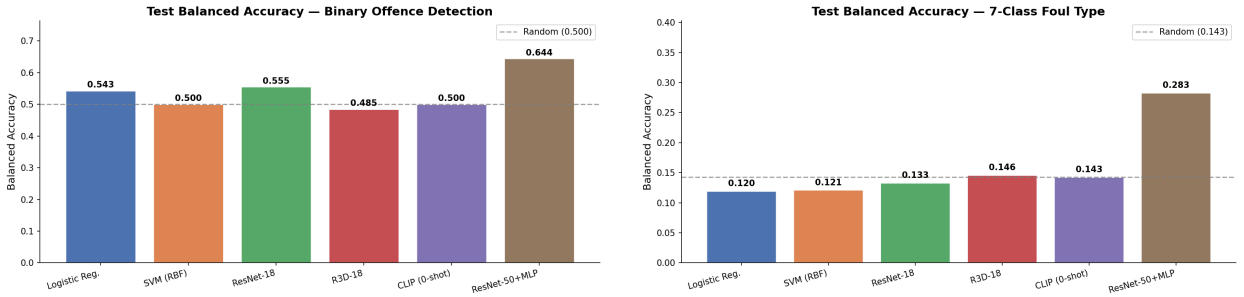
All models were evaluated or reported on the official SoccerNet-MVFoul test split, with the ResNet-50 + MLP caveat described above. The primary metric throughout is *balanced accuracy*, the macro-average of per-class recall. Balanced accuracy is appropriate for imbalanced classification because it gives equal weight to every class. For the binary task the random baseline is 0.500; for the seven-way multiclass task it is $1/7 \approx 0.143$. Per-class F1 scores and confusion matrices are reported as secondary diagnostics to expose degenerate majority-class predictors that would otherwise look superficially competitive under raw accuracy.

4.2 Overall Performance

Table 1 summarises balanced accuracy for all six models on both tasks. Figure 2 visualises the same results as bar charts against the random-guessing baseline.

Table 1: Balanced accuracy for all six models on both tasks. The ResNet-50 + MLP pipeline is the strongest model on both tasks, with cross-validation scores of 0.577 for binary offence detection and 0.266 for seven-way foul-type classification. Its reported test scores are higher, but should be interpreted with the methodological caveat described in Section 3.

Model	**Binary BA**	**Multiclass BA**
Random baseline	0.500	0.143
Logistic Regression	0.543	0.120
SVM (RBF)	0.500	0.121
ResNet-18 (fine-tuned)	0.555	0.133
R3D-18 (fine-tuned)	0.485	0.146
CLIP (zero-shot)	0.500	0.143
ResNet-50 + MLP	**0.644**	**0.283**



(a) Binary offence detection.

(b) Multiclass foul-type classification.

Figure 2: Balanced accuracy for all six models on both tasks. The dashed line indicates the random-guessing baseline. The ResNet-50 + MLP pipeline improves substantially over the best alternative on the binary task and nearly doubles the strongest alternative on the seven-class task.

Two observations stand out. First, **most baselines perform at or below the random-guessing baseline on the binary task**: SVM and CLIP match the 0.500 floor exactly, R3D-18 falls below it, and Logistic Regression and ResNet-18 improve on it only marginally. Second, the ResNet-50 + MLP pipeline achieves 0.644 binary BA and 0.283 multiclass BA, outperforming the best alternative by 8.9 percentage points on the binary task and nearly doubling the strongest alternative on the seven-class task. The gap is especially striking on the seven-class task, where all other models cluster close to the random baseline of 0.143.

4.3 Confusion-Matrix Analysis of Degenerate Predictors

The per-task confusion matrices in Figure 3 reveal that several of the baselines are not merely weak but are producing *degenerate* predictions.

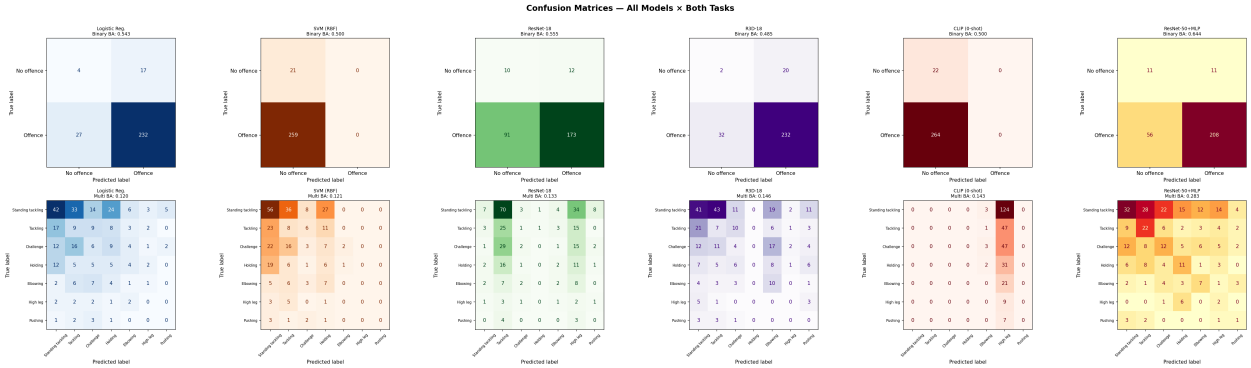


Figure 3: Confusion matrices for all six models on both tasks (top row: binary; bottom row: multiclass). Degenerate predictors are immediately visible: SVM collapses to "No offence" on the binary task, CLIP collapses to "Offence" on the binary task and to "High leg" on the multiclass task, and ResNet-50 + MLP is the only model with non-zero F1 across all seven multiclass categories.

On the binary task, the SVM predicts "No offence" for all 280 test samples used by the handcrafted-feature pipeline. Its 0.500 balanced accuracy is therefore not evidence of useful discrimination, but an artefact of averaging perfect recall on one class with zero recall on the other. CLIP produces the mirror-image failure, predicting "Offence" for all 286 binary test samples. These two models provide no usable binary decision signal despite their apparent proximity to the random baseline. Logistic Regression and ResNet-18 make non-trivial predictions but are still biased toward a dominant class. In contrast, ResNet-50 + MLP recovers 50% of the minority "No offence" samples while maintaining 79% recall on the majority "Offence" class.

The multiclass confusion matrices tell a similar story at higher resolution. The baseline models concentrate predictions into a small number of columns: Logistic Regression and SVM into "standing tackling", "tackling", and "holding"; ResNet-18 into "tackling" and "high leg"; R3D-18 into "standing tackling", "tackling", and "elbowing"; and CLIP almost entirely into "high leg". Only ResNet-50 + MLP places non-zero predicted mass in every column and recovers non-trivial diagonal entries for all seven classes. The weakest class remains "pushing", which is correctly predicted only once out of seven test samples, unsurprising given that pushing is the rarest class and represents less than 3% of the training data.

4.4 Per-Class F1 Analysis

Figure 4 shows per-class F1 scores on the multiclass task. Two patterns are visible. First, baseline methods that achieve non-trivial F1 on "standing tackling" often pay for it with zero F1 on several rare classes; their near-baseline multiclass balanced accuracy is driven mostly by a small subset of the label space. Second, ResNet-50 + MLP is the only model with non-zero F1 on *every* class, including the rare "pushing" class, and is among the strongest models on most individual classes.

Figure 4: Per-class F1 scores on the seven-way foul-type task. Classes are sorted by training frequency from left to right. ResNet-50 + MLP is the only model with non-zero F1 across all seven classes.

4.5 Model-Selection Protocol

The Logistic Regression and SVM hyperparameters were selected using five-fold stratified cross-validation on the training set. The ResNet-50 + MLP hyperparameter configuration was also selected using five-fold stratified cross-validation on cached training-set features. For ResNet-18 and R3D-18, validation-split-based model selection was used instead of k-fold cross-validation because end-to-end retraining for every fold and configuration was prohibitively expensive on the available MPS hardware. CLIP required no training and was evaluated in a zero-shot setting using prompt-based class embeddings.

5 Discussion

5.1 Why Does ResNet-50 + MLP Perform Best?

Three design choices in the pipeline interact favourably on this problem. First, the use of a large *frozen* backbone (ResNet-50) captures the representational advantage of a deeper network without the optimisation cost of end-to-end fine-tuning on limited hardware. The number of trainable parameters is roughly 1–2M for the MLP, compared with tens of millions for an end-to-end CNN. Second, the pipeline averages per-frame features across *all* frames and *all* camera angles before the classifier sees the input, whereas most other models operate on a single middle frame, on a limited number of frames from a single camera, or on per-frame predictions combined by a fixed voting rule. Feature-space averaging of multiple views yields a single denoised representation per action, which is particularly valuable given the fine-grained label space. Third, decoupling the representation from the task makes the cross-validated MLP search tractable on local hardware, while end-to-end fine-tuning of any comparably sized video model would be much more expensive.

5.2 Why Does R3D-18 Fail?

R3D-18 is, in principle, well suited to this task because it explicitly models temporal structure and is pretrained on Kinetics-400. In practice it underperforms every other model on the binary task and falls only slightly above random on the multiclass task. Two factors may help explain this result. First, R3D-18 fine-tuning on MPS is much slower per epoch than ResNet-18, and the feasible search budget on local hardware may have been too limited for stable 3D-CNN fine-tuning. Second, R3D-18 operates on 112×112 inputs rather than 224×224; for a task that turns on fine-grained body-contact cues, the loss of spatial detail may matter more than the added temporal context. The broader lesson is that on this hardware, larger pretrained 2D backbones used as frozen feature extractors are a better use of compute than end-to-end fine-tuning of smaller 3D CNNs.

5.3 Why Do CLIP and the Handcrafted Baselines Collapse?

CLIP's zero-shot performance is weak on both tasks despite multiple prompts per class, two voting schemes, and aggregation across every frame and view. The multiclass confusion matrix shows extreme bias toward the "high leg" class, which accumulates the bulk of predictions regardless of the true label. CLIP's semantic representation is well suited to coarse, visually distinct categories but poorly suited to fine-grained sports actions that differ primarily in transient contact cues. This result suggests that zero-shot vision-language transfer is not a substitute for task-specific supervision in this benchmark.

The SVM's binary prediction collapses to the "No offence" class despite the use of balanced class weights, which reweights the loss but does not change the separability of the RBF feature space. It is likely that the handcrafted features (HOG + optical flow + BoVW) do not separate the two classes in their concatenated feature space, and with sufficiently overlapping feature distributions the class-weighted SVM can still reduce to a constant predictor. Logistic Regression's softer decision boundary permits non-trivial predictions even when the signal is weak, which is why it edges slightly above the random baseline.

6 Limitations and Future Work

This study is limited in several ways that bear directly on how the results should be interpreted. First, the strongest ResNet-50 + MLP result is not a fully unbiased held-out test estimate because the final retraining stage used the test split for model-state selection after cross-validation had selected the hyperparameter configuration. The result is therefore best interpreted as an exploratory upper-bound representation result showing the value of multi-view multi-frame frozen CNN features. A cleaner follow-up experiment should use the official validation split for final model-state selection and evaluate exactly once on the test split.

Second, the binary evaluation is not perfectly identical across all models. The handcrafted-feature models were evaluated on 280 binary test samples because they were derived from the multiclass-filtered feature set, whereas the deep models used 286 binary test samples from the full binary-labelled test set. This discrepancy is small but should be removed in future work by enforcing a shared action-id set across all binary experiments.

Third, all experiments were run on a single Apple M4 MacBook Air with 16 GB of unified memory using PyTorch's MPS backend. This hardware is adequate for frozen-backbone pipelines but prohibitively slow for end-to-end training of large video models, which is why R3D-18 was included in the study as a cautious negative result rather than a fully optimised video-model benchmark.

Several directions would meaningfully extend this work. First, replacing mean-pooling over frames and views with a small attention module, such as a learned transformer-style pooling layer over per-view per-frame features, would allow the model to weight informative views more heavily than long shots. Second, a proper fine-tuning study of R3D-18 on CUDA hardware with a longer epoch budget and a broader learning-rate sweep would establish whether the 3D-CNN disadvantage observed here is intrinsic or an artefact of the local-MPS training regime. Third, combining CLIP image features with the frozen ResNet-50 features before the MLP would test whether vision-language pretraining provides complementary signal even when it is poor on its own. Finally, the dataset's auxiliary annotations (*Severity*, *Body part*, *Contact*) are not used in this project; multi-task learning over these heads could plausibly regularise the foul-type classifier and improve rare-class recall, which remains the weakest aspect of the best model.

7 Conclusion

This project presented a comparison of six machine-learning approaches on the SoccerNet-MVFoul foul-classification problem, evaluated under a balanced-accuracy protocol on two tasks: binary offence detection and seven-way foul-type classification. A multi-view multi-frame pipeline built on a frozen ResNet-50 backbone and a small MLP classifier achieves the strongest reported performance, with 0.644 binary balanced accuracy and 0.283 multiclass balanced accuracy. This substantially improves over the best alternative on the binary task and nearly doubles the strongest alternative on the seven-class task, although the ResNet-50 + MLP result should be interpreted with the test-set model-selection caveat described above.

Confusion-matrix analysis shows that several seemingly competitive baselines collapse into near-constant predictors, which underscores the importance of reporting balanced accuracy together with per-class diagnostics on imbalanced sports-video benchmarks. The principal practical lesson is that, under tight compute constraints, a large frozen-backbone feature extractor combined with multi-view multi-frame feature aggregation and a small cross-validated classifier head is a substantially more effective use of resources than end-to-end fine-tuning of smaller architectures.

Code Availability

All code for data download, feature extraction, model training, and evaluation is available at https://github.com/Shreepoorna1903/soccer-foul-classification. The repository includes a `README.md` with step-by-step instructions for reproducing every result in this report. Dataset access requires a signed SoccerNet NDA.

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